Comparative Analysis of Attention Mechanisms for Automatic Modulation Classification in Radio Frequency Signals

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***Abstract*—Automatic Modulation Classification (AMC) is one of the crucial components in next-generation networks, particularly cognitive radio systems and spectrum management tasks. This study provides a comparative analysis of three attention mechanisms, namely baseline multi-head attention, causal attention, and sparse attention, integrated with Convolutional Neural Networks (CNNs) for radio frequency (RF) signal classification. It proposes a hybrid architecture combining CNN and Transformer models. This architecture uses different attention patterns to capture temporal dependencies in I/Q samples from the dataset, consisting of 11 modulations (8 digital and 3 analog) generated under simulated interference conditions. The experimental results indicate that the baseline attention achieves the highest accuracy of 85.05%. On the other hand, causal and sparse attention mechanisms deliver significant computational advantages, reducing inference time by 83% and 75%, respectively, and show classification performance above 84%. In addition, distinct attention patterns for different modulation schemes provide the design of efficient attention mechanisms for real-time radio signal processing tasks.**

***Index Terms*—Automatic Modulation Classification, Attention Mechanisms, Deep Learning, Radio Frequency, Convolutional Neural Networks, Transformer Architecture**

I. Introduction

Automatic Modulation Classification (AMC) is essential in next-generation networks (5G and beyond), particularly cognitive radio systems and spectrum management tasks. Enabling cognitive radios allows adaptively configuring transmission parameters, while spectrum management systems allow monitoring electromagnetic environments [1], [2]. Traditional feature-based approaches in communication systems heavily rely on statistical methods, which suffer from generalization under different channel conditions. Recent advances in the artificial intelligence (AI) landscape, particularly deep learning, have significantly revolutionized AMC approaches, with Convolutional Neural Networks (CNNs) demonstrating better performance in extracting spatial features from I/Q signal representations [3]. However, radio signals possess temporal dependencies due to the nature of communication systems, which makes it difficult to capture effectively for CNNs. The attention mechanisms, particularly in the Transformer architecture, have demonstrated significant success in modeling long-range dependencies for various domains.

In recent years, AMC has seen significant advancements with the adoption of deep learning techniques in modern communication systems. In the literature, one of the early studies introducing deep learning for AMC is the use of CNNs, as proposed in [4], which demonstrated the capability of CNNs in extracting local features from complex-valued radio signals. Building on this, lightweight and efficient models have been proposed, such as Teng et al.'s polar feature-based model with online channel compensation [5]. The authors in [6] explore the use of deep neural networks (DNNs) for AMC by extracting 21 statistical features from received signals (BPSK, QPSK, 8PSK, 16QAM, 64QAM). Results indicated that DNNs can offer significant performance improvements compared to traditional classifiers, especially in high Doppler fading conditions. The authors in [7] also investigate the security vulnerabilities of AI-based AMC systems against adversarial attacks, and show these systems offer low performance under such attacks, and mitigation strategies, e.g., defensive distillation, can effectively enhance model robustness against adversarial threats.

With advanced communication and computing technologies, recent focus has shifted toward combining spatial and temporal representations. For instance, Lin et al. [8] introduced a time-frequency attention mechanism within a CNN architecture, enabling selective feature emphasis in both domains. Similarly, Zhang et al. [9] proposed a ResNet enhanced with involution layers, serving as an efficient self-attention mechanism for signal classification. The study [10] proposes a deep learning-based AMC scheme combining random erasing and an attention mechanism to enhance performance. At the first step, two data augmentation methods, i.e., random erasing at the sample level and at the amplitude/phase (AP) channel level, are introduced to improve model robustness and generalization. Then, a signal embedding containing modulation information and a single-layer LSTM with an attention module are used to better capture temporal features of modulated signals. Experimental results on the RML2016.10a dataset demonstrate that the proposed

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approach achieves competitive results compared to state-of-the-art methods.

Transformer-based architectures have also gained traction in AMC. Cai et al. [11] implemented a pure Transformer encoder, achieving superior accuracy, particularly under low SNR conditions. MobileRaT [12], a lightweight Transformer variant, was introduced for drone communications, balancing accuracy and resource constraints. The authors in [13] introduce a novel deep neural network designed for AMC, called MCformer. It effectively captures temporal correlations in signal embeddings by combining convolutional layers with self-attention-based encoder layers. According to results, MCformer offers a satisfactory classification accuracy at all signal-to-noise ratios on the RadioML2016.10b dataset, while using significantly less parameters, making it suitable for fast and energy-efficient applications. Similarly, another group of authors in [14] proposes PCTNet, which is a novel parallel CNN-transformer network for AMC. PCTNet combines the local feature extraction of CNNs with the long-range dependency modeling of transformers through a parallel architecture. The results demonstrate that PCTNet provides better classification accuracy compared to traditional deep learning models. Hybrid models further advance the field. Wang et al. [15] combined CNN, Transformer, and graph neural networks in their CTGNet model to capture both spatial and topological dependencies. A multimodal attention-based MLP architecture was proposed in [16], which fuses time-domain and spectrogram features for robust AMC under noisy environments. These studies highlight the growing trend toward attention-driven and Transformer-inspired models in AMC, justifying this study's comparative focus on attention variants within a CNN-Transformer combination.

This study examines an essential question: *How do different attention mechanisms affect the performance and computational efficiency of AMC systems?* To examine this question, the following three distinct attention patterns are investigated:

- **Baseline Multi-Head Attention**: Basic self-attention mechanism allowing bidirectional information flow
- **Causal Attention**: Masked attention mechanism enforcing temporal causality constraints
- **Sparse Attention**: Local windowed attention mechanism reducing computational complexity

The main contributions of this paper include: (1) A novel hybrid architecture combining CNN and Transformer models, particularly designed for radio signal classification; (2) A comparative analysis of attention mechanisms on the RML2016.10a dataset, which consists of 11 modulations (8 digital and 3 analog); (3) Detailed computational efficiency analysis indicating significant speed-ups, and (4) Modulation performance analysis for attention pattern selections.

The source code is also available at the GitHub repository[1].

[1]https://github.com/ocatak/attention_based_automatic_modulation_recognition

II. Methodology

A. Problem Formulation

Figure 1 shows the overall system overview. Given a radio frequency signal represented as complex-valued I/Q samples $\mathbf{x} = [x_1, x_2, \ldots, x_T] \in \mathbb{C}^T$, where T is the sequence length, the objective is to classify the modulation scheme $m \in \mathcal{M} = \{m_1, m_2, \ldots, m_K\}$ where K is the number of modulation classes.

We reformulate the complex signal as a real-valued tensor $\mathbf{X} \in \mathbb{R}^{2\times T}$ where:

$$\mathbf{X} = \begin{bmatrix} \mathrm{Re}(x_1) & \mathrm{Re}(x_2) & \cdots & \mathrm{Re}(x_T) \\ \mathrm{Im}(x_1) & \mathrm{Im}(x_2) & \cdots & \mathrm{Im}(x_T) \end{bmatrix} \tag{1}$$

B. CNN-Transformer Hybrid Architecture

The proposed CNN-Transformer hybrid architecture combines two capabilities: (1) spatial feature extraction capabilities from CNNs and (2) the temporal modeling power from attention mechanisms. The architecture consists of three main components:

1) CNN Feature Extractor: The CNN backbone extracts hierarchical features from I/Q samples:

$$\mathbf{H}^{(1)} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv1D}(\mathbf{X}; 32, 7))) \tag{2}$$

$$\mathbf{H}^{(2)} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv1D}(\mathbf{H}^{(1)}; d_{model}, 5, 2))) \tag{3}$$

where $\mathrm{Conv1D}(input; filters, kernel, stride)$ denotes a 1D convolution operation, while BN represents batch normalization, and $d_{model} = 64$ is the model dimension.

2) Multi-Head Attention Module: Given the CNN features $\mathbf{H} \in \mathbb{R}^{B\times L\times d_{model}}$ where B is batch size and L is sequence length, we apply multi-head attention:

$$\mathrm{MultiHead}(\mathbf{H}) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)\mathbf{W}^O \tag{4}$$

$$\mathrm{head}_i = \mathrm{Attention}(\mathbf{H}\mathbf{W}_i^Q, \mathbf{H}\mathbf{W}_i^K, \mathbf{H}\mathbf{W}_i^V) \tag{5}$$

where $\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V \in \mathbb{R}^{d_{model}\times d_k}$ are learnable projection matrices, $d_k = d_{model}/h$, and $h = 4$ is the number of attention heads.

3) Attention Variants: In this study, three distinct attention mechanisms are examined, and each mechanism has its own unique characteristics and computational properties as follows:

Baseline Multi-Head Attention: This approach presents the standard attention mechanism used in the original Transformer architecture, i.e., standard scaled dot-product attention mechanism. It allows unrestricted bidirectional information flow between all sequence positions:

$$\mathrm{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{6}$$

The attention weights $\mathbf{A}_{ij} = \mathrm{softmax}(\mathbf{Q}_i\mathbf{K}_j^T/\sqrt{d_k})$ represent the position j when processing position i. This mechanism has quadratic complexity $O(L^2)$ where L is the sequence length, as each position can attend to all other positions.

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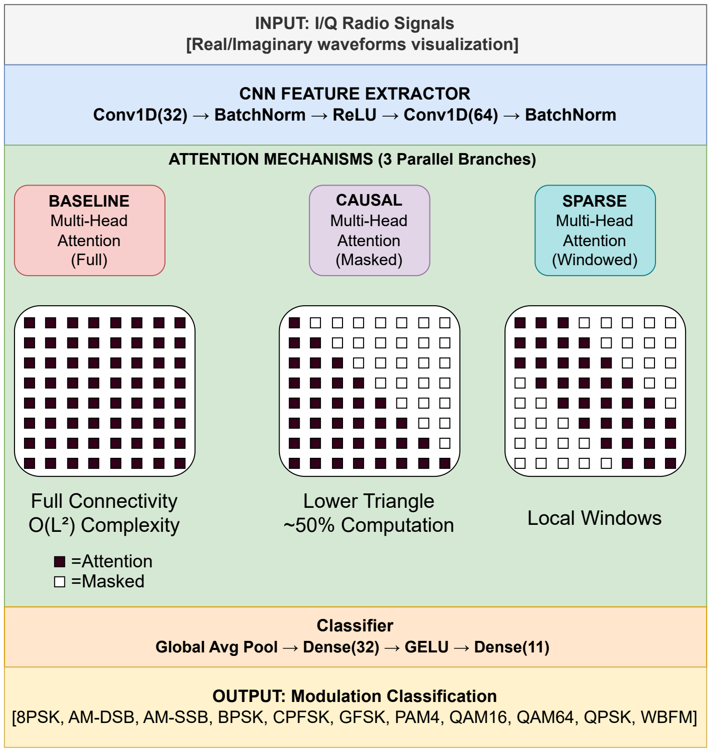

Fig. 1: System Overview

The global receptive field enables modeling of long-range dependencies, but at a high computational cost.

Causal Attention: This approach is inspired by autoregressive language models; causal attention enforces temporal causality by restricting attention to previous and current positions only:

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T + \mathbf{M}_{causal}}{\sqrt{d_k}}\right)\mathbf{V} \quad (7)$$

where $\mathbf{M}_{causal}$ is a lower triangular mask:

$$\mathbf{M}_{causal}[i,j] = \begin{cases} 0 & \text{if } j \le i \\ -\infty & \text{if } j > i \end{cases} \quad (8)$$

This constraint prevents possible unintended information exposure from future time steps, where only past and current samples are available. In this mechanism, $O(L^2)$ complexity is maintained while the masked operations are reduced by approximately 50%.

Sparse Attention: This approach aims to address the quadratic complexity limitation. Therefore, a local windowed sparse attention mechanism is utilized to limit attention to a fixed-size neighborhood:

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T + \mathbf{M}_{sparse}}{\sqrt{d_k}}\right)\mathbf{V} \quad (9)$$

where the sparse mask $\mathbf{M}_{sparse}$ is given as follows:

$$\mathbf{M}_{sparse}[i,j] = \begin{cases} 0 & \text{if } |i-j| \le w/2 \\ -\infty & \text{otherwise} \end{cases} \quad (10)$$

Where the window size (w) is set to eight (8) to reduce complexity to $O(L{\cdot}w)$. This provides significant computational savings while maintaining local temporal modeling capabilities.

Each attention mechanism illustrates a different trend between modeling capacity and computational cost/efficiency:

- **Baseline**: Offer maximum expressivity while highest computational cost
- **Causal**: Enable real-time compatibility with moderate computational savings
- **Sparse**: Focus on local modeling and offer maximum computational efficiency

III. Experimental Setup

A. Dataset

The proposed approach is evaluated on the RML2016.10a dataset, generated with GNU Radio, for AMC research. The dataset contains 220,000 I/Q samples for 11 modulation schemes: 8PSK, AM-DSB, AM-SSB, BPSK, CPFSK, GFSK, PAM4, QAM16, QAM64, QPSK, and WBFM. Each sample consists of 128 complex-valued points captured at varying signal-to-noise ratio (SNR) levels ranging from -20dB to 18dB.

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Algorithm 1 Multi-Head Attention Mechanisms

Input: Input features $\mathbf{H} \in \mathbb{R}^{B\times L\times d_{model}}$, attention type $\tau \in$ {baseline, causal, sparse}
Output: Output features $\mathbf{O} \in \mathbb{R}^{B\times L\times d_{model}}$
// Multi-head projection
for $i = 1$ to h **do**
  $\mathbf{Q}_i \leftarrow \mathbf{H}\mathbf{W}_i^Q$ {Query projection}
  $\mathbf{K}_i \leftarrow \mathbf{H}\mathbf{W}_i^K$ {Key projection}
  $\mathbf{V}_i \leftarrow \mathbf{H}\mathbf{W}_i^V$ {Value projection}
end for
// Compute attention scores
for $i = 1$ to h **do**
  $\mathbf{S}_i \leftarrow \frac{\mathbf{Q}_i\mathbf{K}_i^T}{\sqrt{d_k}}$ {Scaled dot-product}
  // Apply attention-specific masking
  if $\tau =$ causal **then**
    for $j = 1$ to L, $k = 1$ to L **do**
      if $k > j$ **then**
        $\mathbf{S}_i[j, k] \leftarrow -\infty$ {Mask future positions}
      end if
    end for
  else if $\tau =$ sparse **then**
    for $j = 1$ to L, $k = 1$ to L **do**
      if $|j - k| > w/2$ **then**
        $\mathbf{S}_i[j, k] \leftarrow -\infty$ {Mask distant positions}
      end if
    end for
  end if
  $\mathbf{A}_i \leftarrow \text{softmax}(\mathbf{S}_i)$ {Attention weights}
  $\mathbf{A}_i \leftarrow \text{Dropout}(\mathbf{A}_i, p)$ {Attention dropout}
  $\text{head}_i \leftarrow \mathbf{A}_i\mathbf{V}_i$ {Weighted values}
end for
// Concatenate and project heads
$\mathbf{C} \leftarrow \text{Concat}(\text{head}_1, \ldots, \text{head}_h)$
$\mathbf{O} \leftarrow \mathbf{C}\mathbf{W}^O$ {Output projection}
return $\mathbf{O}$

Algorithm 2 CNN-Transformer Training Algorithm

Input: Dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, Model θ, Learning rate α, Attention type τ
Output: Trained model parameters θ^*
Initialize parameters θ randomly
Split $\mathcal{D}$ into train $\mathcal{D}_{train}$, validation $\mathcal{D}_{val}$, test $\mathcal{D}_{test}$
for epoch $= 1$ to E_{max} **do**
  total_loss $\leftarrow 0$
  for batch $\mathcal{B} \subset \mathcal{D}_{train}$ **do**
    // Forward pass
    $\mathbf{X} \leftarrow \text{Preprocess}(\mathcal{B})$ {I/Q normalization}
    $\mathbf{H} \leftarrow \text{CNNExtractor}(\mathbf{X})$ {Feature extraction}
    // Transformer blocks
    for $l = 1$ to N_{layers} **do**
      $\mathbf{A}^{(l)} \leftarrow \text{MultiHeadAttention}(\mathbf{H}^{(l-1)}, \tau)$ {Alg. 1}
      $\mathbf{H}^{(l)} \leftarrow \text{LayerNorm}(\mathbf{H}^{(l-1)} + \text{Dropout}(\mathbf{A}^{(l)}))$
      $\mathbf{F}^{(l)} \leftarrow \text{FFN}(\mathbf{H}^{(l)})$ {Feed-forward network}
      $\mathbf{H}^{(l)} \leftarrow \text{LayerNorm}(\mathbf{H}^{(l)} + \text{Dropout}(\mathbf{F}^{(l)}))$
    end for
    $\mathbf{z} \leftarrow \text{GlobalAvgPool}(\mathbf{H}^{(N_{layers})})$ {Sequence pooling}
    $\hat{\mathbf{y}} \leftarrow \text{Classifier}(\mathbf{z})$ {Final prediction}
    // Backward pass
    $\mathcal{L} \leftarrow \text{CrossEntropy}(\hat{\mathbf{y}}, \mathbf{y})$
    total_loss $\leftarrow$ total_loss $+ \mathcal{L}$
    $\theta \leftarrow \theta - \alpha\nabla_\theta\mathcal{L}$ {Parameter update}
  end for
  // Validation and early stopping
  val_acc $\leftarrow \text{Evaluate}(\mathcal{D}_{val}, \theta)$
  if Early stopping criteria met **then**
    break
  end if
  Update learning rate schedule
end for
return θ^*

For experiments, samples within the SNR range of -6dB to 18dB are filtered to focus on practical operating conditions. The dataset is split into three groups, i.e., training (60%), validation (20%), and testing (20%) sets, respectively.

B. Implementation Details

In this study, all models are implemented in PyTorch. Adam with Decoupled Weight Decay optimizer (AdamW) is set with an initial learning rate of 10^{-3}, weight decay of 10^{-4}, and cosine annealing scheduling. Training also uses early stopping with patience of 15 epochs based on validation accuracy.

Data augmentation includes Gaussian noise injection with 30% probability and a standard deviation of 0.02. Each sample is normalized per instance to have zero mean and unit variance.

The model architecture consists of:

- CNN: 2 convolutional layers (32, 64 filters)
- Transformer: 2 layers, 4 attention heads, 256-dimensional feed-forward network (FFN)
- Classifier: 2-layer Multilayer Perceptron (MLP) with GELU activation and dropout rate of 0.1

C. Evaluation Metrics

The models are evaluated using the following performance metrics:

- **Accuracy**: Measure the overall model classification accuracy by making the correct prediction
- **F1-Score**: Utilize the harmonic mean of precision and recall
- **Inference Time**: The use of average forward pass time per sample
- **Parameters**: Define the total number of trainable parameters

IV. Results and Analysis

A. Overall Performance Comparison

Table I summarizes the performance of all three attention mechanisms. The baseline model achieves the highest test accuracy of 85.05%, while causal and sparse attention mechanisms show a competitive performance, i.e., above 83%, less than 85%

TABLE I: Overall Performance Comparison

Model	Test Acc. (%)	Avg F1	Params (M)	Inf. Time (ms)
Baseline	85.05	0.843 ± 0.129	0.11	0.06
Causal	83.93	0.832 ± 0.133	0.11	0.02
Sparse	83.64	0.830 ± 0.136	0.11	0.03

B. Computational Efficiency Analysis

Figure 2 shows the significant computational advantages of causal and sparse attention mechanisms compared to the baseline attention mechanism. Causal attention achieves an 83% reduction in inference time (0.02ms vs 0.12ms), while sparse attention achieves a 75% reduction (0.03ms vs 0.12ms). These results also indicate that causal and sparse attention

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mechanisms are more suitable for real-time applications in which latency is critical.

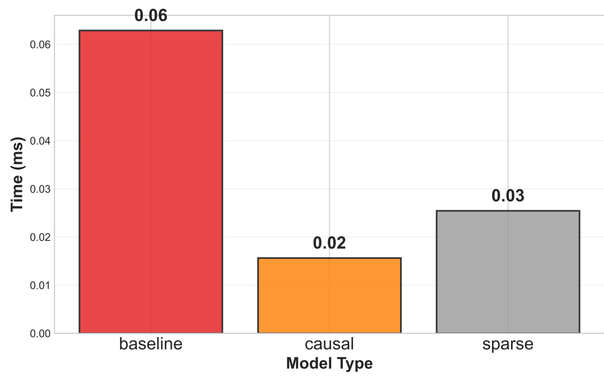

Fig. 2: Inference time comparison across attention mechanisms

C. Per-Modulation Analysis

Table II illustrates the detailed performance metrics for each modulation method in terms of F-1 Score. Several interesting patterns are observed:

Modulations with Baseline Attention: All modulations, except AM-DSB and WBFM, with baseline attention mechanism offer a satisfactory performance in terms of F1-Score, ranging from 0.803 to 0.953.

Simple Modulations Excel with Sparse Attention: CPFSK, GFSK, and PAM4 achieve the best performance for sparse attention (0.938, 0.934, and 0.984, respectively), indicating that local temporal patterns are sufficient for these modulations.

Complex Modulations Prefer Full Attention: QAM16 and QAM64 modulations demonstrate low performance for restricted attention patterns, i.e., causal and sparse. This indicates the importance of global context for these modulations.

WBFM Challenges: All attention mechanisms struggle with WBFM classification (F1 $\approx$ 0.5), probably due to its fundamentally different spectral characteristics compared to digital modulations.

TABLE II: Per-Modulation F1-Score Comparison

Modulation	Baseline	Causal	Sparse
8PSK	0.860	**0.869**	0.851
AM-DSB	**0.736**	0.733	0.732
AM-SSB	**0.910**	0.898	0.905
BPSK	**0.953**	0.950	0.947
CPFSK	0.928	0.916	**0.938**
GFSK	0.915	0.927	**0.934**
PAM4	0.979	0.983	**0.984**
QAM16	**0.803**	0.741	0.727
QAM64	**0.810**	0.745	0.718
QPSK	0.887	**0.892**	0.885
WBFM	0.496	0.504	**0.509**

D. Attention Pattern Visualization

Figure 3 shows the F1-score distribution for each attention mechanism. The boxplots display that baseline attention has slightly higher median performance compared to causal and sparse attention mechanisms.

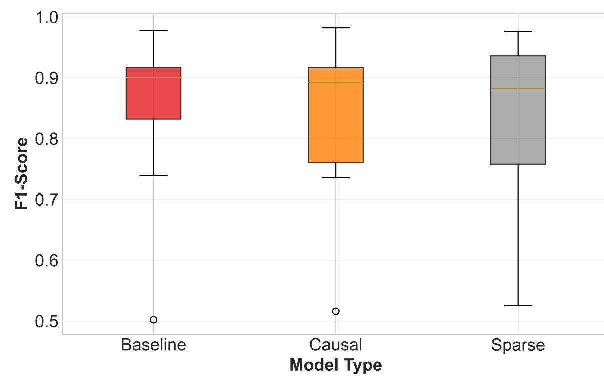

Fig. 3: F1-Score distribution by attention mechanism

E. Confusion Matrix Analysis

The confusion matrices (Figures 4) show consistent error patterns across all models. The observations are as follows:

QAM Confusion: QAM16 and QAM64 show mutual confusion, reflecting their similar constellation structures with different symbol densities.

AM-DSB/WBFM Confusion: These analog modulations show cross-confusion, suggesting shared spectral characteristics that challenge all attention mechanisms.

Digital Modulation Robustness: BPSK, QPSK, and PSK achieve high classification accuracy for all attention mechanisms.

V. Discussion and Future Work

A. Attention Mechanism Insights

The results demonstrate that the specific requirements of the application should guide the choice of attention mechanism:

Real-time Applications: Causal and sparse attention provide excellent accuracy and computational efficiency, making them ideal for real-time spectrum monitoring and cognitive radio applications.

High-accuracy Requirements: Baseline attention can still be the best choice when maximum classification accuracy is prioritized over computational efficiency.

Mixed Deployment: A hybrid approach could be an option for dynamically selecting attention mechanisms based on detected signal characteristics.

B. Architectural Considerations

The proposed hybrid architecture effectively combines spatial and temporal modeling capabilities from CNN and Transformer models. The CNN model provides spatial feature extraction, while attention mechanisms from the Transformer model enable the capture of long-range dependencies for modulation classification in modern communication systems.

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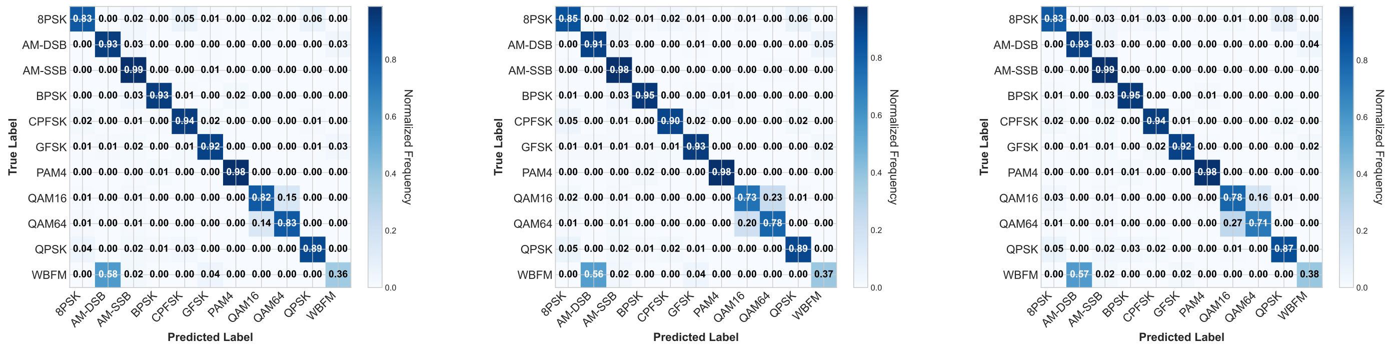

(a) Baseline Model (b) Causal Model (c) Sparse Model

Fig. 4: Confusion matrices for different attention mechanisms

C. Limitations and Future Directions

Several limitations warrant future investigation:

- **SNR Sensitivity**: Analysis for different SNR ranges could indicate the robustness of the attention mechanism to noise.
- **Channel Effects**: Evaluation under realistic channel conditions (fading, multipath) would enhance practical relevance.
- **Adaptive Attention**: Dynamic attention pattern selection based on signal characteristics could optimize performance-efficiency trade-offs.
- **Larger Datasets**: Evaluation on larger, more diverse datasets would strengthen generalization claims.

VI. Conclusion

This study provides a comprehensive comparative analysis of attention mechanisms for automatic modulation classification (ACM), which is essential in modern communication systems. CNN-Transformer hybrid architecture with three attention variants, i.e., baseline, causal, and sparse, was evaluated on the RML2016.10a dataset, consisting of 11 modulations (8 digital and 3 analog) at varying signal-to-noise ratios. According to the results, key findings are: (1) Baseline attention achieves highest accuracy (85.05%), but with significant computational cost, (2) Causal and sparse attention provide 83% and 75% inference time reductions, respectively while maintaining competitive performance above 83%, (3) Simple modulations benefit from sparse local attention, while complex modulations require global context, and (4) All mechanisms struggle with analog modulations, particularly WBFM.

The results are expected to provide useful guidance for engineers and researchers developing efficient attention mechanism architectures for next-generation communication networks (5G and beyond) based on accuracy and computational efficiency requirements.

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